CS23334 Fundamentals of Data Science Lab

Experiment 4

To Count the frequency of occurrence of a word in a body of text is often needed during text processing.

Description: Import the word_tokenize function and gutenberg.

Code:

```
import nltk

from nltk.tokenize import word_tokenize

from nltk.corpus import gutenberg

nltk.download('gutenberg')

nltk.download('punkt')

sample = gutenberg.raw("austen-emma.txt")

token = word_tokenize(sample)

wlist = []

for i in range(50):
    wlist.append(token[i])

wordfreq = [wlist.count(w) for w in wlist]

print("Pairs\n" + str(list(zip(wlist, wordfreq))))
```

OUTPUT:

```
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816',
1), (']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma',
2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever',
1), (',', 5), ('and', 3), ('rich', 1), (',', 5), ('with', 2), ('a', 1),
('comfortable', 1), ('home', 1), ('and', 3), ('happy', 1), ('disposi-
tion', 1), (',', 5), ('seemed', 1), ('to', 1), ('unite', 1), ('some', 1),
('of', 2), ('the', 2), ('best', 1), ('blessings', 1), ('of', 2), ('e-
xistence', 1), (';', 1), ('and', 3), ('had', 1), ('lived', 1), ('nearly
', 1), ('twenty-one', 1), ('years', 1), ('in', 1), ('the', 2), ('world'
, 1), ('with', 2)]
```

Result:

Word frequency analysis was successfully performed on Jane Austen's 'Emma' using NLTK. Common stopwords like 'the', 'to', and 'and' appeared most frequently in the first 50 tokens, demonstrating basic NLP tokenization and frequency counting techniques.